

The Saboteur

Every AI tool answers your STEM questions. We built the one that makes you audit the AI's work.

The 2026 problem

AI now usually gives *correct* answers — and that's the danger, not the cure. The failure mode isn't that AI is wrong, it's that fluent, confident, mostly-correct output trains students to stop checking. Over half of STEM students rely on intuition rather than systematic checks to detect AI errors. A better model makes this worse, not better. Existing AI-literacy tools are PDFs and curricula — *reading material*, not practice. No consumer tool trains a student to catch a wrong AI answer on calibrated mistakes, with a measured improvement signal.

What we built

A single-player web game that flips the role: instead of asking an AI, you **audit** it. Each round shows a worked algebra solution — either fully correct, or containing exactly one planted error. The player decides Trust or Flag-the-step. The system grades instantly against hidden ground truth.

The architectural moat

SymPy decides correctness. The LLM only narrates. Every problem is generated symbolically; errors are injected at known step indices using a misconception library of 15 transforms, each unit-tested to produce verifiably-wrong but well-formed output. The grader is deterministic. The LLM phrases steps and explains the misconception after the reveal, behind prompts that forbid correctness claims and a post-filter that strips evaluative language. The system *cannot* be confidently wrong about correctness — by construction.

The calibration metric (the killer number)

	Solution CORRECT	Solution WRONG
You trusted	✓ Correct trust (+5)	✗ Over-trust (-10)
You flagged	✗ Over-suspicion (-5)	✓ Correct catch (+10, +5 bonus)

Over-trust is weighted 3× over-suspicion, directly encoding the documented dangerous failure mode. The session score climbs as players improve — verifier-backed, not a proxy.

The Phase-3 differentiator: Audit Real AI Answers

Paste any LLM's step-by-step algebra solution into the BYOAI tab. SymPy verifies each step against the previous one and tells you exactly where the AI broke. The game becomes a daily tool — for real homework, real ChatGPT conversations.

Architecture

- Problem Generator (SymPy) → canonical solution + per-step operation tags
- Sabotage Engine → injects 1 error at a known step from the Misconception Library
- LLM Layer → narrates each step and explains the misconception (NEVER judges correctness)
- Frontend (React + Tailwind + KaTeX) → Trust/Flag interaction, calibration dashboard
- Grader + Calibration Engine → 2×2 outcomes, Elo-tuned difficulty

Stack

Python, SymPy, FastAPI, SQLite, React 18, Vite, Tailwind, KaTeX, Recharts, and LLM.API (sponsor).

Scope discipline

Deep tool, narrow domain. Algebra only: linear (1-step, distribution, fraction) and quadratic (factoring, formula, perfect square). Six generators, 15 misconceptions, 200+ verified instances. We don't claim universal STEM coverage; we claim the domain we cover is deterministically verified and the misconceptions mirror documented student/AI failure modes.